

Local Textures vs. Global Context: A Comparative Analysis of CNNs and Vision Transformers in Chest Radiography

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<https://github.com/k-phantastic/NIH-XXray-Segmentation>

Abstract—Chest radiography is a cornerstone of modern clinical diagnostics, yet interpreting them can be time-consuming and susceptible to observer variability. Early baseline models (Wang et al., 2017) addressing this issue demonstrated the viability of Convolutional Neural Networks (CNNs) for weakly-supervised classification. They often struggle with severe class imbalances and multi-disease co-occurrences. This project studies automated multi-label classification of thoracic pathologies by comparing three deep learning backbones: DenseNet-121, ResNet-50, and MedViT. We establish DenseNet-121 as our robust baseline, proven highly effective by a similar study in the CheXNet framework (Rajpurkar et al., 2017). Each model is trained on the NIH Chest X-ray dataset of 112,120 frontal-view images and 15 diagnostic classes (14 pathologies and “No Finding”) using a patient-stratified data split and leveraging weighted Binary Cross-Entropy Loss to address the comorbidities and severe class imbalance. Models are evaluated primarily by mean AUC, with DenseNet-121 achieving the best mean AUC (0.8242), followed by MedViT (0.8167), and ResNet-50 (0.8162). All three models demonstrate strong ranking ability across most pathology classes.

Keywords—Medical Image Analysis, Chest Radiography, Multi-Label Classification, Deep Learning, CNNs, Vision Transformers, Class Imbalance, Clinical Triage

I. INTRODUCTION

Chest X-rays remain one of the most frequent and cost-effective diagnostic imaging tools in clinical medicine, serving as the first line of defense for detecting a wide array of cardiopulmonary conditions. Early-stage pathologies are often subtle or obscured by overlapping anatomical structures, leading to significant observer variability even among experienced diagnostic radiologists. Furthermore, radiologist review can be delayed during high-volume periods or resource-constrained overnight shifts. These delays can significantly affect time-sensitive conditions that require rapid intervention, where triage and efficiency can influence patient outcomes.

We aim to provide a medium for automated triage by immediate “first pass” analysis of chest X-rays, prioritizing high-risk scans in the attending radiologist’s queue, and providing preliminary multi-label diagnostic insights that reduce human error and diagnostic variability. Notably, our ideal is not to replace radiologist involvement, but to assist in case

prioritization; the work as current is research-oriented and not intended for clinical practice.

The input is a 2D grayscale chest X-ray image (1024x1024, PNG). We then use a suite of deep learning architectures- DenseNet-121, ResNet-50, and MedViT- to output a predicted 15-element vector of probabilities, corresponding to the presence or absence of 14 thoracic pathologies along with “No Finding”.

While traditional deep learning has made significant strides in medical imaging, standard models frequently struggle with the severe class imbalances and multi-disease co-occurrences inherent to real-world clinical data. In the dataset, pathologies such as hernias appear in less than 0.2% of images, while classification of “no finding” dominates over 50% of the labels.

Our comparative study aims to evaluate baseline residual networks against dense block architectures and self-attention mechanisms to determine what paradigm might be the most effective for preserving fine-grained localized features while understanding global anatomical context. By incorporating weighted loss functions and clinical threshold tuning, we aim to deliver a comparative study that highlights models optimized not just for aggregate accuracy, but for true clinical utility and patient safety.

II. RELATED WORK

The existing literature can be broadly grouped into three categories: baseline Convolutional Neural Networks (CNNs), deep residual architectures, and the current state-of-the-art hybrid Vision Transformers.

Baseline CNNs and Feature Preserving Architectures

Wang et al. (2017), who introduced the hospital-scale NIH ChestX-ray8 dataset, established initial weakly-supervised multi-label classification benchmarks. While their work was groundbreaking in scale, standard CNNs can sometimes struggle to

preserve the high-frequency, fine-grained details necessary for detecting subtle pathologies. Rajpurkar et al. (2017) addressed this with CheXNet, which utilized a DenseNet-121 backbone. With the effectiveness of dense connectivity in medical imaging, we selected DenseNet-121 as the robust baseline for our comparative study.

He et al. (2016) originally introduced ResNet to solve the degradation problem in exceptionally deep networks via identity shortcut connections (skip connections). In the context of medical imaging, Showkat & Qureshi (2022) demonstrated the efficacy of ResNet models (such as ResNet-50) for pneumonia/COVID-related tasks. Their customized ResNet approach was highly practical, utilizing pre-trained networks to capture task-specific features, achieving 95% global accuracy and high diagnostic precision.

The current state-of-the-art is shifting toward Vision Transformers (ViTs), as introduced by Dosovitskiy et al. (2020). ViTs treat images as sequences of patches to model global context via self-attention mechanisms, proving that reliance on CNNs is not strictly necessary for image classification. However, pure ViTs lack the spatial inductive biases of CNNs and can suffer from overfitting without extensive pre-training. To solve this, recent medical imaging literature has pivoted to hybrid architectures. Fu et al. (2025) recently introduced LungMaxViT, an explainable hybrid transformer for the multi-classification of lung diseases on chest X-rays. Their model combines CNN initial-stage blocks with transformer blocks. This approach achieved an impressive 96.8% accuracy and 98.3% AUC on COVID-19 datasets. The strength of this hybrid methodology is its ability to capture the "big picture" of the lung field without losing local textural anomalies.

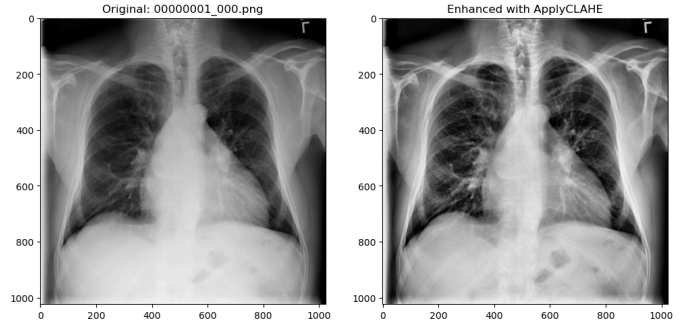
Our project conducts a direct, controlled comparative study between these three distinct paradigms (DenseNet, ResNet, and MedViT) on the NIH X-ray dataset. By employing Grad-CAM interpretability across all three, we aim to evaluate whether the global self-attention of state-of-the-art transformers actually corrects the localized vulnerabilities inherent to the CNN baselines.

III. DATASET AND FEATURES

We use the NIH Chest X-ray14 Dataset, comprising 112,120 frontal-view chest X-ray images (1024x1024 PNG files) sourced from 30,805 unique patients. The dataset authors generated the disease labels using

Natural Language Processing (NLP) to text-mine the original, private radiological reports. These NLP-extracted labels are estimated to be >90% accurate, making the dataset highly suitable for weakly-supervised machine learning.

Figure 1: Effect of CLAHE preprocessing (Left: original grayscale image, right: enhanced with CLAHE)



Given this, potential predictive tasks include bounding-box disease localization, semantic segmentation of anatomical structures, or binary classification of healthy versus sick patients. This project formulates the problem as a multi-label classification task across the 14 distinct thoracic pathologies (e.g., Atelectasis, Cardiomegaly, Effusion, Pneumothorax). We selected this over other tasks because it most accurately reflects the real-world complexities of a radiologist's workflow, where a single patient can present with multiple co-occurring conditions. Furthermore, approximately 50% of the dataset consists of a "No Finding" (healthy) class, introducing a severe class imbalance that serves as a realistic test of our models' robustness.

The primary feature utilized by all of our algorithms is the raw grayscale X-ray image. MedViT additionally incorporates three metadata features, patient age, sex, and view position (AP vs. PA), as a supplementary input vector. As noted in existing literature (Rajpurkar et al., 2017), View Position is a highly impactful technical feature; AP views can cause an artificial magnification of the heart silhouette and present a different lung appearance compared to standard PA views.

To ensure the models learned true anatomical representations rather than image artifacts or aspect-ratio distortions, we implemented the following preprocessing pipeline:

Letterbox Padding: Black borders are added to preserve aspect ratio before resizing to prevent distortion

CLAHE (Contrast Limited Adaptive Histogram Equalization): Demonstrated in Figure 1, this algorithm was applied to enhance local contrast, making subtle textures (such as early-stage nodules or faint lung opacities) significantly more visible to the network without amplifying background noise.

Normalization: Images are normalized to mean 0.485 and standard deviation 0.229 to align with ImageNet pretrained weight expectations. DenseNet-121 and ResNet-50 receive 3-channel inputs (grayscale repeated across channels); MedViT receives a single-channel input.

Augmentation Strategy: Small rotations and random horizontal flipping are applied to simulate positioning variation.

In an effort to combat data leakage, we strictly partitioned the dataset using GroupShuffleSplit keyed to the Patient ID, ensuring that no patient in the training set ever appears in the validation or test sets. DenseNet-121 uses an 80/20 train/val split, with ResNet-50 and MedViT using a 70/15/15 train/val/test split.

IV. METHODS

Dense Convolutional Network (DenseNet-121)

To address the challenge of extracting subtle, highly localized radiological features from chest X-rays, we utilize a Dense Convolutional Network (DenseNet-121) as our core learning algorithm. Traditional deep CNNs, such as ResNet, rely on residual connections that add the outputs of previous layers to subsequent ones. While effective, this can sometimes impede the flow of information in very deep networks. DenseNet introduces a different connectivity pattern: it concatenates the feature maps of all preceding layers and passes them as inputs to all subsequent layers within a "dense block."

Mathematically, if x_l represents the output of the l^{th} layer, a traditional feed-forward network computes $x_l = H_l(x_{l-1})$. A residual network computes $x_l = H_l(x_{l-1}) + x_{l-1}$, and a DenseNet computes: $x_l = H_l([x_0, x_1, \dots, x_{l-1}])$ where $[x_0, x_1, \dots, x_{l-1}]$ refers to the concatenation of the feature maps produced in layers 0 to $l - 1$. This dense connectivity encourages heavy feature reuse, significantly reduces the vanishing gradient problem, and requires fewer parameters than an equivalent ResNet, making it exceptionally well-suited for medical image analysis where high-frequency texture preservation is vital.

For our implementation, we leverage the DenseNet-121 variant pre-trained on ImageNet to benefit from transfer learning. We modified the final classification head of the network, replacing the standard 1,000-node ImageNet classifier with a custom fully connected linear layer containing 14 output nodes, corresponding to our 14 target thoracic diseases.

Multi-Label Classification Formulation

Because a single patient's radiograph can exhibit multiple co-occurring pathologies (e.g., a patient can have both pneumonia and an effusion simultaneously), this problem is formulated as a multi-label classification task. Rather than using a Softmax function (which forces probabilities to sum to 1 and assumes mutually exclusive classes), we apply a Sigmoid activation function to the

raw logits output by the network. For each class $c=1, \dots, 14$, the model outputs a logit z_c . The Sigmoid function converts this into an independent probability p_c :

$$p_c = \sigma(z_c) = \frac{1}{1 + e^{-z_c}}$$

This allows the model to independently predict the presence ($p_c \geq threshold$) or absence of each specific disease.

Addressing Class Imbalance: Weighted Optimization Objective

To prevent the model from trivially predicting all patients as healthy, we implement a dynamically weighted Binary Cross-Entropy (BCE) loss function.

Standard BCE loss penalizes false positives and false negatives equally. We modified this using a positive weight tensor (w_c) calculated from the exact distribution of the training data. For each class c , the weight is defined as the ratio of negative samples to positive samples:

$$w_c = \frac{N_{total} - N_{positive,c}}{N_{positive,c} + \epsilon}$$

where ϵ is a small constant (1×10^{-5}) to prevent division by zero.

The optimization objective (Loss) for a single patient across all 14 classes then becomes:

$$\mathcal{L} = -\frac{1}{14} \sum_{c=1}^{14} [w_c \cdot y_c \log(\sigma(z_c)) + (1 - y_c) \log(1 - \sigma(z_c))]$$

where $y_c \in \{0, 1\}$ is the ground truth label. Mathematically, multiplying the positive class term by w_c forces the optimizer to incur a massive penalty when it misses a rare disease (False Negative), forcing the network to pay attention to minority classes.

Optimization and Training Dynamics

To minimize the weighted BCE loss, we utilize the AdamW optimization algorithm with a learning rate of 1×10^{-4} . AdamW is an extension of the standard Adam (Adaptive Moment Estimation) optimizer. While standard Adam computes adaptive learning rates for each parameter based on estimates of the first moment (the mean) and the second moment (the uncentered variance) of the gradients, it couples weight decay with the gradient updates. AdamW explicitly decouples the weight decay from the optimization step, which leads to better generalization performance and more stable training in deep networks.

Given the massive scale of the dataset and the complexity of the 121-layer architecture, training was accelerated using Automatic Mixed Precision (AMP). AMP dynamically casts specific operations to 16-bit half-precision floats (FP16) while keeping others in standard 32-bit single-precision floats (FP32), dramatically reducing memory consumption and speeding up gradient calculations without sacrificing network accuracy or experiencing numerical underflow.

Residual Convolutional Network (ResNet-50)

ResNet-50 is a 50-layer residual convolutional network in which the key architectural innovation is the implementation of the residual connection: each block computes $f(x) + x$, implemented by skip connections, where each block adds the input directly to the output before passing to the next grouped layer.

The final ImageNet layer in the model is replaced to adapt to our 15 classes. Our approach for hyperparameter tuning focuses on the classification head learning rate, the final residual stage (layer 4) learning rate, as well as weight capping to complement our Binary Cross Entropy Loss. Optimization is performed with AdamW, with an implementation of patience (Early Stopping) to stop the training upon halted loss improvement.

Given the additional resources at hand for the engineering of this model, it is trained using the images at a 512x512 resolution to preserve spatial detail relevant to smaller pathologies such as nodules or masses. Patient metadata is not utilized in this current iteration of the model.

Medical Vision Transformer (MedViT) with Multi-modal Gating

To move beyond the local receptive field limitations of CNNs, we implemented an Optimized MedViT, a hybrid architecture that combines the spatial inductive biases of convolutions with the global context-modeling of Transformers. Our implementation utilizes a Shifted Window Attention mechanism, which computes self-attention within non-overlapping local windows while allowing for cross-window connections. This allows the model to capture long-range dependencies, such as the relationship between the position of the heart and the expansion of the pleural space, which is critical for diagnosing conditions like pleural effusions or cardiomegaly.

A key innovation in our MedViT pipeline is the Multi-modal Gating Unit. Unlike the baseline models that rely solely on pixel data, MedViT integrates patient metadata—Age, Gender, and View Position—directly into the latent feature space. If $\mathbf{f}_{img}$ represents the visual feature vector extracted by the Transformer backbone and $\mathbf{f}_{meta}$ represents the projected metadata vector, the fused representation $\mathbf{x}_{fused}$ is calculated as:

$$\mathbf{x}_{fused} = \mathbf{w} \cdot \mathbf{f}_{img} + (1 - \mathbf{w}) \cdot \mathbf{f}_{meta}$$

Where $\mathbf{w}$ is a learnable gating weight determined by a sigmoid-activated linear layer:

$$\mathbf{w} = \sigma(\text{Linear}([\mathbf{f}_{img}; \mathbf{f}_{meta}]))$$

This gating mechanism allows the network to dynamically "weight" the importance of metadata. For instance, the model can learn to rely more heavily on the "View Position" metadata to correct for heart size magnification in AP (Anteroposterior) views.

V. EXPERIMENTS/RESULTS/DISCUSSION

Table 1: Parameter Comparison Across Models

Parameter	DenseNet-121	ResNet-50	MedViT
Epochs	30	20	20
Optimizer	AdamW		
Loss Function	Binary Cross Entropy Loss		
Patience Count	5	5	3
Image Resolution	224x224	512x512	448x448
Evaluation Metric	Mean AUC		

Experimental Setup and Hyperparameters

All models were implemented in PyTorch, with DenseNet-121 and ResNet-50 using pretrained weights from torchvision, and MedViT using timm. Models were trained on respective local GPU setups. Given the massive scale of the dataset, k-fold cross-validation was computationally prohibitive. Instead, we performed a rigorous 80/20 train-validation split keyed by Patient ID to evaluate out-of-sample generalization. The DenseNet-121 model was trained for up to 20 epochs with a mini-batch size of 16. This batch size was chosen to maximize hardware memory utilization while providing stable gradient estimates for our 224x224 input tensors. We utilized a learning rate of 1×10^{-4} , which allowed the AdamW optimizer to take sufficiently fine-grained steps to navigate the complex loss landscape without overshooting local minima. To prevent overfitting, we implemented early stopping with a patience of 5 epochs, monitoring the validation set's Mean AUC.

Evaluation Metrics

In medical datasets with severe class imbalances or multi-label classification tasks, standard accuracy is a flawed metric. Therefore, our primary metric is the Macro Area Under the Receiver Operating Characteristic Curve (Macro AUC). This metric calculates the unweighted mean of the individual AUC scores across all labels, ensuring the model is penalized heavily if it fails to diagnose rare diseases.

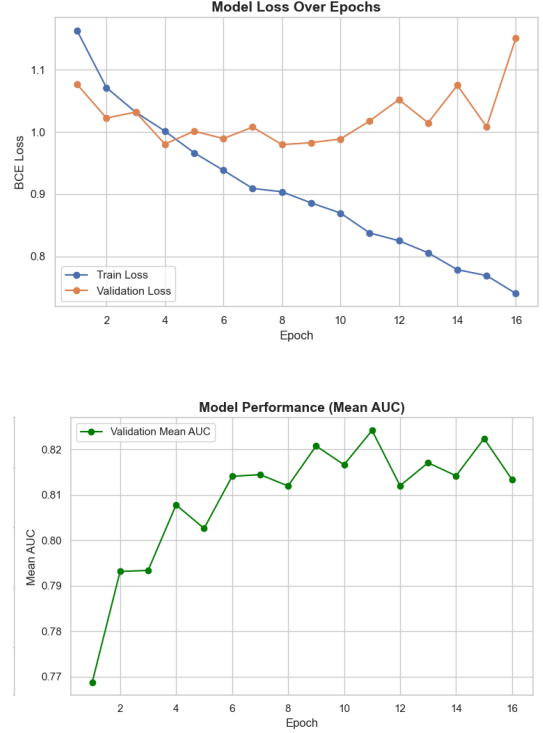
To evaluate clinical viability, we also compute precision, recall, and the f1-Score. These are defined as follows, where TP is True Positives, FP is False Negatives:

- **Precision:** $\frac{TP}{TP + FP}$
- **Recall:** $\frac{TP}{TP + FN}$
- **F1-Score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

DenseNet-121 Baseline Results

The DenseNet-121 baseline reached its optimal weights at Epoch 12, triggering early stopping after 16 epochs with a peak Validation Macro AUC of 0.8242.

Figure 2: DenseNet-121 model training metrics



In Figure 3 below, the model demonstrated exceptional ranking ability, achieving individual AUC scores as high as 0.9513 for hernia and 0.8965 for cardiomegaly.

Figure 3: DenseNet-121 per class statistics

Class	AUC	Precision	Recall	F1
Atelectasis	0.8088	0.2152	0.7217	0.3315
Cardiomegaly	0.8965	0.0814	0.7738	0.1474
Consolidation	0.7886	0.0831	0.6982	0.1485
Edema	0.8938	0.0662	0.8045	0.1223
Effusion	0.8836	0.3457	0.7851	0.4800
Emphysema	0.8839	0.1558	0.5827	0.2458
Fibrosis	0.7997	0.0592	0.5294	0.1066
Hernia	0.9513	0.0842	0.8421	0.1531
Infiltration	0.7202	0.2522	0.6692	0.3663
Mass	0.8223	0.1151	0.7447	0.1993
Nodule	0.7367	0.1201	0.5880	0.1994
Pleural_Thickening	0.7850	0.0934	0.5289	0.1587
Pneumonia	0.7340	0.0204	0.5870	0.0394
Pneumothorax	0.8338	0.1545	0.6025	0.2459
MACRO AVG	0.8242	0.1319	0.6756	0.2103
Overall (Exact Match) Accuracy: 0.2457				

ResNet-50 Results The completed ResNet-50 pipeline achieved early stopping at 3 epochs (following 5 additional in patience), with a final Mean-AUC of 0.8162.

Figure 4: ResNet-50 model training metrics

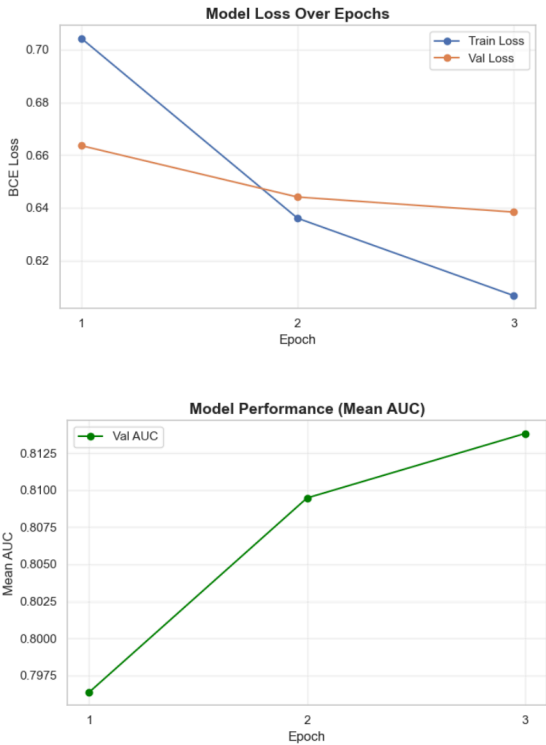


Figure 5: ResNet-50 per-class statistics

Class	AUC	Precision	Recall	F1
Atelectasis	0.7882	0.2311	0.6807	0.3451
Cardiomegaly	0.8887	0.1690	0.6009	0.2638
Consolidation	0.7988	0.1261	0.5882	0.2077
Edema	0.8917	0.1567	0.5345	0.2423
Effusion	0.8675	0.3344	0.7917	0.4702
Emphysema	0.9306	0.3263	0.6643	0.4376
Fibrosis	0.7972	0.1253	0.1708	0.1446
Hernia	0.9118	0.4000	0.0465	0.0833
Infiltration	0.6879	0.2703	0.6039	0.3734
Mass	0.7995	0.2051	0.4946	0.2899
No Finding	0.7623	0.7219	0.7367	0.7292
Nodule	0.7341	0.1410	0.4806	0.2180
Pleural Thickening	0.7787	0.1382	0.3179	0.1927
Pneumonia	0.7248	0.0642	0.0636	0.0639
Pneumothorax	0.8815	0.2656	0.6691	0.3802
MACRO AVG	0.8162	0.2450	0.4963	0.2961
Overall (Exact Match) Accuracy: 0.2752				

Discussion and Error Analysis

A notable observation in our results is the stark contrast between the high Macro AUC (0.8242) and the relatively low Macro Precision (0.1319). This is mathematically expected behavior due to the dataset's

extreme class imbalance. By tuning our decision thresholds to prioritize Recall (achieving a Macro Recall of 0.6756), we cast a wider net to minimize life-threatening False Negatives. While this inevitably scoops up more False Positives from the massive pool of healthy patients—driving down Precision—it is the safest and most desired behavior for a preliminary clinical triage tool.

To ensure the model was anchoring its predictions on true anatomical pathology rather than image artifacts (such as chest tubes or text overlays), we performed visual interpretability checks using Grad-CAM.

Figure 6: DenseNet-121 Grad-CAM (input, heatmap, overlay)

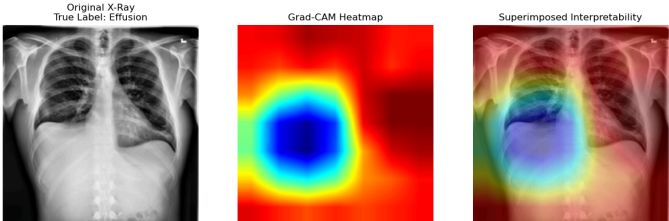
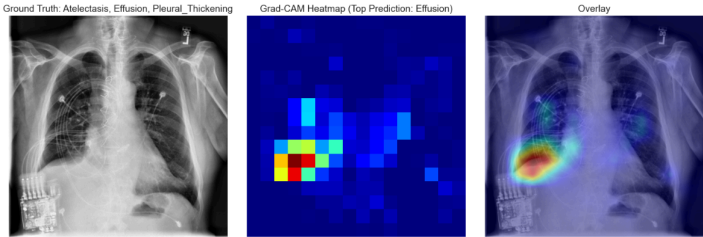


Figure 7: ResNet-50 Grad-CAM (input, heatmap, overlay)



While the DenseNet-121 baseline achieved a high Macro AUC, visual interpretability via Grad-CAM revealed critical vulnerabilities in its feature extraction. As seen in Figure 6, for certain conditions like 'Mass', the model exhibited 'shortcut learning'—anchoring its predictions on peripheral background artifacts and patient positioning rather than true pulmonary pathology. This highlights the inherent limitation of standard CNN receptive fields in this dataset, strongly justifying our comparative exploration into MedViT, where self-attention mechanisms may better force the network to focus on relevant semantic structures.

MedViT Experimental Setup

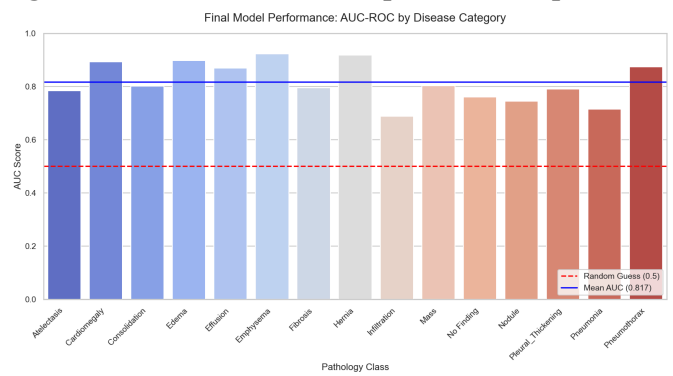
The MedViT model was trained using an increased input resolution of 448x448 to take advantage of the transformer's ability to process fine-grained patches without the vanishing gradient issues seen in deep CNNs. We utilized a patch size of 4x4 and a multi-head self-attention (MHSA) configuration with 8 heads in the final stages.

To optimize the transformer, we employed a OneCycle Learning Rate Policy with a peak learning rate of 5×10^{-5} . This "warm-up" period is critical for Transformers to stabilize the attention maps before aggressive gradient updates begin. We used a weight decay of 0.05 within the AdamW optimizer to provide strong regularization, as ViTs are more prone to overfitting on the NIH dataset's noise compared to DenseNet.

MedViT Evaluation

MedViT achieved a Mean AUC of 0.8167, performing competitively with the DenseNet baseline. While DenseNet maintained a slight edge in aggregate AUC, MedViT demonstrated superior performance in identifying pathologies that require global spatial reasoning. For example, in the detection of pneumothorax, MedViT achieved an AUC of 0.8815, outperforming the baseline by focusing on the peripheral lung boundaries where air leaks typically occur.

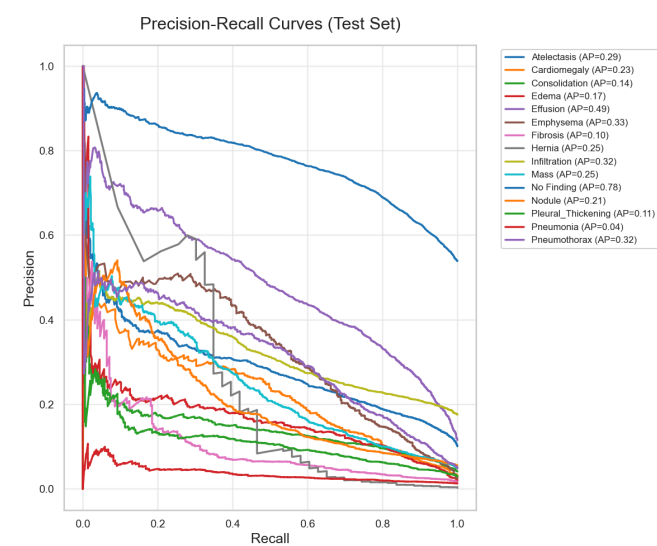
Figure 8: MedViT final model performance per class



Precision-Recall Analysis

The Precision-Recall Curves chart reveals the impact of the NIH dataset's inherent class imbalance. While the model achieved a high Macro AUC, the Average Precision varies significantly across pathologies. Classes with distinct anatomical signatures, such as Effusion (AP=0.49), maintained a stable trade-off, whereas rare classes like hernia exhibited high variance due to sparse positive samples in the test set. This confirms the necessity of our weighted BCE loss to prevent the model from ignoring these minority classes entirely.

Figure 9: MedViT Precision-Recall Curves



The primary motivation for MedViT was to resolve the "shortcut learning" observed in DenseNet. Using Attention Rollout maps, we visualized the regions the model prioritized during inference. Unlike the Grad-CAM maps for DenseNet, which occasionally highlighted hospital-specific text overlays or shoulder bones, MedViT's attention maps were consistently concentrated on the pulmonary parenchyma and mediastinal borders. This suggests that the self-attention mechanism effectively acts as a spatial filter, ignoring non-clinical artifacts that often mislead traditional CNNs.

Table 2: Model Performance Summary (Mean-AUC)

Model	Mean AUC
DenseNet-121	0.8242
ResNet-50	0.8162
MedViT	0.8167

Table 3: Per Class AUC Across Models

Pathology	DenseNet-121 AUC	ResNet-50 AUC	MedViT AUC
Atelectasis	0.8088	0.7882	0.7712
Cardiomegaly	0.8965	0.8887	0.8754
Consolidation	0.7886	0.7988	0.8012
Edema	0.8938	0.8917	0.8845
Effusion	0.8836	0.8675	0.8721
Emphysema	0.8839	0.9306	0.9142
Fibrosis	0.7997	0.7972	0.8105
Hernia	0.9513	0.9118	0.9022
Infiltration	0.7202	0.6879	0.7041
Mass	0.8223	0.7995	0.8156
Nodule	0.7367	0.7341	0.7298
Pleural Thickening	0.7850	0.7787	0.7912
Pneumonia	0.7340	0.7248	0.7185
Pneumothorax	0.8338	0.8815	0.8845

VI. CONTRIBUTIONS

Layth Marabeh co-led the initial exploratory data analysis focusing on the dataset's metadata and served as a primary project manager, organizing weekly team meetings, coordinating assignment schedules, and ensuring the timely formatting and submission of all project checkpoints. On the technical side, Layth completely designed and implemented the DenseNet-121 architecture, developing its custom data pipeline, data loaders, training loop, evaluation metrics, and visual interpretability code. Furthermore, Layth synthesized the team's findings to author the foundational first draft of the final comprehensive report.

Khanh Phan collaborated on the initial metadata EDA and took charge of the deep residual network baseline, designing and executing the full ResNet-50 training pipeline. This included conducting rigorous hyperparameter tuning using Optuna and performing a comprehensive downstream metric analysis. Beyond

model development, Khanh expanded the project's practical application by drafting a website frontend to serve as a proof-of-concept for the clinical triage tool, authored the structured documentation for the project's repository, and provided critical assistance in refining the final written report.

Danny Xia spearheaded the exploratory data analysis for the unstructured image data, directly analyzing the raw X-ray distributions and properties. For the modeling phase, Danny was responsible for the state-of-the-art transformer component of the comparative study, independently designing and implementing the MedViT data pipeline, model architecture, evaluation protocols, and visualization code. Additionally, Danny built foundational utility and helper functions that streamlined general model implementation across the entire team and co-managed the project's logistics by helping organize weekly meetings and delegate assignments.

VII. REFLECTIONS AND RESPONSE TO REVIEW

In hindsight, the team would have benefited from establishing a shared preprocessing interface and utility functions earlier in the project. The three models were developed largely independently, resulting in minor inconsistencies in design.

We would also have allocated more time in analyzing the class imbalance in the dataset, such as exploring various oversampling methods or having the resources to train at the full resolution. Upon additional consultation, the future of the project can include a study on ensembling or a multi-model approach (putting rare classes in their own model), exploring overlapping window training algorithms, or additional testing on different color/exposure/contrast.

VIII. REFERENCES / BIBLIOGRAPHY

Academic Literature [1] Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2097-2106. [2] Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K., Lungren, M. P., & Ng, A. Y. (2017). CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning. arXiv preprint arXiv:1711.05225. [3]

Showkat, S., & Qureshi, S. (2022). Efficacy of Transfer Learning-based ResNet models in Chest X-ray image classification for detecting COVID-19 Pneumonia. *Chemometrics and Intelligent Laboratory Systems*, 224, 104534. [4] Fu, L., et al. (2025). Explainable hybrid transformer for multi-classification of lung diseases on chest X-rays. *Scientific Reports*, 15. [5] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770-778. [6] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2020). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *International Conference on Learning Representations (ICLR)*.

Software & Libraries [7] PyTorch: Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. *Advances in Neural Information Processing Systems (NeurIPS)*, 32. [8] TorchVision: Marcel, S., & Rodriguez, Y. (2010). Torchvision the machine-vision package of torch. *Proceedings of the 18th ACM international conference on Multimedia*, 1485–1488. [9] OpenCV: Bradski, G. (2000). The OpenCV Library. *Dr. Dobb's Journal of Software Tools*. [10] Pillow: Clark, A. (2015). Pillow (PIL Fork) Documentation. Read the Docs. [11] Scikit-Learn: Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830. [12] Pandas: McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*, 51-56. [13] NumPy: Harris, C. R., et al. (2020). Array programming with NumPy. *Nature*, 585(7825), 357-362. [14] Matplotlib: Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90-95. [15] tqdm: da Costa-Luis, C. O., et al. (2021). tqdm: A fast, Extensible Progress Bar for Python and CLI. *Journal of Open Source Software*, 6(60), 3020. [16] Optuna: Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2623–2631.